

Representation Convergence - Status at a Glance

All models pretrained and frozen; only closed-form linear maps fitted, except rows marked (trained). Thesis: independently trained models share representations up to a linear transformation - different CHARTS of one shared relational manifold - and that convergence is deployable. Every number held-out, 1,000-item galleries throughout (chance R@1 0.001). Model roster, metric interpretation, the GPT-2 case, local-vs-global and the shape figures: Technical_Cheat_Sheet.pdf A11-A15.

Test	Row title	Models	Data / split (rows per dim)	Question	Result
A-B · THE CORE PROOF Two experiments establish that convergence is linear and deployable: an LLM pair with nothing in common but language, and a vision pair shipped as a phone adapter.					
A	Two LLMs converge	GPT-2 vs Pythia-160M	10k WikiText passages, text only; 7,500 fit / 2,500 eval (9.8)	Do two independently trained LLMs share representations?	Yes, linearly. Stitching R2 0.70-0.80 mid-layers, peak 0.797 (L11) vs 0.173 random; CKA 6.3x control; trained curve rises with depth, random decays - opposite slopes
B	Vision adapter works	MobileCLIP-S1 -> SigLIP 2	4k COCO val2017 images, encoded twice; 3,000 fit / 1,000 eval (5.9)	Are two vision encoders linearly interchangeable?	Ridge R2 0.592 / cosine 0.900 ; Procrustes 0.038 -> anisotropy (69x knob spread); MLP +0.003 -> exactly linear
B3/D	Deployed and measured	adapter + shared Qdrant index	no fitting; 1,000-image gallery, first captions as queries	Is the convergence usable in production?	0.79 MB fp16 adapter, one shared index: R@5 95.3% of ceiling (raw = chance); verif AUC 0.999 / rank 0.987 / membership 0.889; hubness 4.28 -> 1.61
E · CROSS-MODAL, SCALE, GEOMETRY Nothing is trained. A vision encoder that never saw text is mapped into text spaces to test whether convergence crosses modality, grows with capacity, and depends on the target's geometry.					
E1	Crosses modality	DINOv2 -> bge-m3	8k COCO train2017 image-caption pairs, 5 captions averaged; 7,000 / 1,000 (9.1); ceiling re-measured	Does convergence cross modalities without joint training?	Yes, measurably: R@1 0.358 = 40.9% of matched ceiling, 358x chance; robust across caption protocols
E1 sweep	Scales with capacity	DINOv2 small/base/large -> bge	9.5k / 18k / 23.8k pairs, N scaled with width so 11.1 rows/dim throughout	Does alignment grow with encoder capacity? (PRH)	Yes. 40.9 / 53.5 / 58.3% of ceiling over 13.8x params; monotone in 6 measures; +15.3 pts per decade (fit R2 0.949); compressive
E1.1	Collapsed target fails to read	DINOv2 -> GPT-2 (vs bge)	identical to E1 by construction	Does the text side need similarity training?	R2 retains 71%, retrieval 34%; GPT-2 eff. rank 6.5/768, pair-cos +0.999 - cosine reads 1.000 at 9.5% retrieval
E1.2	Isotropy rescues it	DINOv2 -> GPT-2 + label-free fixes	six corrections, each fitted on train targets only	Is the collapse fixable without labels?	Whitening: R@1 0.130 -> 0.478 = 102.6% of a 1.2B-pair contrastive encoder; replicated 101.4%; iso-retrieval corr +0.94
E1.3	Mechanism predicts	ladder: GPT-2 / BERT / SBERT / bge	two unseen encoders, predictions written before running	Does the C.11 mechanism predict out of sample?	P1 rank-corr +1.00 (confirmed); P2 partial: isotropy necessary, NOT sufficient - residual spread 0.110
F · THE CONTRAST CONDITION The one series that deliberately trains, to test whether capacity or data changes the answer. It does not.					
F1-F2	MLP = linear, twice	SigLIP 2 teacher, MLP head (trained)	fresh 40k corpus, 37,999 / 2,000 (74.2)	Does a non-linear head beat the matrix?	No. cosine 0.9183 = 0.9183, gain +0.0000, on TWO independent runs; R@1 residual after data effect +0.008 / +0.003 - noise
F5	Data helps a little; gate not met	linear W refit on 38k rows	same corpus, 5,000 fresh queries	Does more data lift the linear map?	R@1 +0.0244 (bootstrap CI excludes 0); 90%-of-ceiling gate NOT met at 89.0% - reported as such
F3	Gated shut on purpose	unfreeze MobileCLIP stage	built, not run	Would unfreezing help?	Not run: it would void the 'no pretrained weight modified' statement. The gate is the contribution
C · RECOVERY WITHOUT PAIRS The pairing is withheld entirely; the correspondence must be found from each space's own shape.					
C1.1	Correspondence self-identifies	MobileCLIP <-> SigLIP 2, no pairs	1,500 vectors each, one side shuffled, permutation sealed	Can the pairing be recovered from geometry alone?	Gromov-Wasserstein 39.7% exact = 595x chance ; map from it R@1 0.252, AUC 0.906. Negatives: ICP degrades; whitening kills matching (->0.1%). Scope: both encoders contrastive
G · THE SHARED HUB One 512-d linear coordinate system that all encoders enter. Tests whether it costs anything, whether trained parts move between encoders, across lineages, and at what scale structure is shared.					
G1	One hub, seven encoders	DINOv2 x3, bge, GPT-2, BERT, SBERT	9,533 shared images; 8,533 / 1,000; width x alpha swept	Is there ONE shared coordinate system?	Transfer R@1 0.451 (AUC 0.986), cliff at 512->768; unchanged vs 5 spaces (within noise). Composition 95.6% on 36/41 pairs
G1	Read-only spaces	same	same	Can the hub write back into every space?	GPT-2: source 95.1%, target 24.8% - readable out of, not writable into. Cause tested: target items at +0.999, ranking needs precision an extra stage can't keep. Reach does NOT predict this; eff. rank does
G3	Shared beats bespoke	same 7	per-pair hubs as the fair baseline	Does one space cost anything vs per-pair hubs?	42/42 pairs SHARED , mean retention 103.6%; control at chance; GPT-2 rescued 4x by hub whitening
G4	Crosses lineage	SigLIP 2 into the FROZEN hub	same 9,533 prefix; same head, unchanged	Does transfer survive an unrelated lineage?	94.2% of native - inside the 93.8-96.5% within-family band; control at chance. Still a ViT; ConvNeXt is the open test
G5	Agreement is LOCAL	all 21 pairs	kNN overlap at k=1..500, 1,500 items, ratio to chance	Is the shared structure local or global?	Falls monotonically for every pair: 967x at k=1 -> 2.3x at k=500 . Objective (+53x) > lineage (+44x) > modality (+29x), exploratory
DEMOS · THE SYSTEM WORKING Interactive notebooks that show the results in use rather than as tables. Not evaluations - galleries are widened for coverage, so quote R@1 only from the 1,000-item runs.					
B5	Phone path, simulated	MobileCLIP + fp16 adapter	10 images per gallery; fp16 exactly as Core ML computes it	Does the exact phone pipeline retrieve correctly?	Free-text and compositional queries, negative controls, fp16/fp32 parity check - all pass
B6	Two tiers, one index	server gallery + user upload	upload indexed via phone path, auto-captioned (BLIP)	Can a phone photo and server text meet in one Qdrant collection?	Free-text search over ONE shared index; top-3 with per-tier probabilities; mobile vs server timing
G2	Live hub, both directions	7-encoder hub, URL / upload / path input	hub is a CPU linear solve; GPU only for the fresh forward pass; ~9.5k-item demo gallery	Image -> captions and text -> images through the hub?	Both work. Text -> images uses hub coordinates only - no head, no readout - the shared space doing the work directly. Hub persists as a 16 MB fp16 file
H / V · CERTIFICATION Not experiments: tooling that checks every artifact before any negative result is believed.					
H1	47 checks, no failures	every stored artifact	power, alignment, sanity, target geometry, cache keys	Are the artifacts sound?	NO hard failures; 2 flags = the documented GPT-2 collapse, reproduced in two files; rows/dim 5.9-9.8; shuffle gaps 0.55-0.86
V1	Independent reproduction	Experiment B pair	separate code path	Do reported numbers reproduce?	verif AUC 0.9994, R@1 0.987, 69x knob spread - three reported figures reproduced

Status at a Glance - page 2: achievements, geometry, defense, restrictions

Achievements - the nine findings

#	Achievement	Headline evidence	Where
1	Convergence is linear	MLP = ridge to 4 decimals, twice	C.9
2	Related, not rigid	Procrustes 0.038 vs ridge 0.592; 69x spread	4
3	Deployable	95% of ceiling @ 0.79 MB; AUC 0.999	4, D.2
4	Crosses modality	40.9-58.3% of ceiling, 358x chance	C.8
5	Scales with capacity	+15.3 pts / decade, 6 monotone	C.10
6	Isotropy, not training	whitening -> 102.6% / 101.4%	C.11
7	Self-identifying	GW 39.7% = 595x chance, 0 pairs	C.12
8	One space, used	93-96% transfer, cross-lineage 94.2%	C.13
9	Mechanism predicts	ladder rank-corr +1.00	C.14

Geometry flow



Geometry flow. Same territory, different projections: nothing shared in raw coordinates, structure shared before any fit, one matrix supplies the frame, one hub serves them all. Agreement is LOCAL (967x -> 2.3x from k=1 to 500); the naive 'same space' claim is refuted; the 'different charts of one manifold' claim is supported.

Defense - hard questions, answered

Objection	Answer, with the number
Isn't whitening cheating?	Label-free, train-targets-only, linear - an instance of the thesis. Whiten o ridge is still one matrix.
Is transfer circular?	Random-map control collapses to chance; absolute R@1 RISES on unseen encoders - a memorising head can't do that.
Hub can't write into GPT-2 - a hole?	Bounded and explained: reads out at 95%; failure is ranking inside a +0.999 space. Deployed use meets IN the hub.
Only local structure - so what?	Measured, stated as a restriction, and it explains why the adapter misses part of the ceiling.
Cross-lineage still shares objective	Yes: SigLIP is contrastive and a ViT. Result is cross-lineage, not cross-objective; ConvNeXt is open.
Why bge if SBERT is better?	Comparability with every ceiling in the report; SBERT (101.0% vs 98.1%) offered as alternative.
Wishful interpretations?	Four explanations TESTED and FALSIFIED, kept in the record: shape-vs-collapse, hub discards directions, reach predicts writability, more encoders help.

Restrictions - scope the claims do not exceed

Claim	Measured scope	NOT claimed
Linear convergence	GPT-2/Pythia; MobileCLIP/SigLIP; DINOv2 x3; 4 text encoders	all architectures / scales
Scales with capacity	DINOv2 family 22M-304M, compressive	beyond 304M; other families
Isotropy rescue	GPT-2 target, 2 image scales	sufficiency (C.14 bounds it)
Pair-free recovery	39.7% @ 595x; both encoders contrastive	a deployable pairing-free system
Shared hub	7 encoders, 4 objectives, COCO, English; cross-lineage 94.2%	universal space for ALL models / architectures
Writing into a space	reads all; writes all but collapsed GPT-2 (24.8%)	that a collapsed space can be reconstructed for ranking
Structure is shared	LOCALLY - 100s x chance at k=1, ~2x at k=500	global metric identity
Domain	one image domain, one language, one caption corpus	open-world scale; other domains